

# Shuo Guo

The Chinese University of Hong Kong, Shenzhen  
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## Education

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**The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)** Shenzhen, China  
B.Sc. in Applied Mathematics Sept 2025 – May 2029 (expected)  
Advisor: Prof. Dr. Zhi-Quan Tom Luo

## Research Interests

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- Conserved quantities in non-convex optimization.
- Geometry of flat minima in deep learning optimization.
- Theoretical optimization and the mathematical foundations of machine learning.
- Dynamics, stability, and convergence of stochastic and adaptive optimization algorithms, especially Adam and Muon, with particular interest in dynamical-systems and stochastic-analysis approaches.

## Experience

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**Research Assistant, School of Artificial Intelligence, CUHK-Shenzhen** Sept. 2026 – present  
Advisor: Prof. Dr. Xiaopeng Li  
Thesis: Conserved Quantities of First Order Method in Non-convex Optimization  
**Research Assistant, CUHK-Shenzhen** May 2026 – Jul 2026  
Worked on the empirical evaluation of optimization methods for neural-network training.

## Honors and Awards

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- Muse Full-tuition Scholarship, CUHK-Shenzhen

## Selected Background

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- **Convex Optimization:** Studied selected topics from *Lectures on Convex Optimization*, including convexity, gradient methods, strong convexity, complexity bounds, and accelerated first-order methods.
- **Deep Learning and Optimization:** Studied training dynamics and the theoretical analysis of adaptive optimization methods, with particular emphasis on the Adam family of algorithms; read selected chapters of *Mathematical Introduction to Deep Learning: Methods, Implementations, and Theory*.
- **Stochastic Analysis:** Studied measure-theoretic probability and stochastic calculus in course *introduction to stochastic analysis* course, including Brownian motion, stochastic integration, stochastic differential equations, and Itô's formula.
- **Data Science:** Studied selected topics from *Foundations of Data Science*, including SVD, random graphs, concentration, and VC dimension, through an independent course led by Prof. Dr. Zhi-Quan Tom Luo.

## Technical Skills

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- **Programming:** Python; numerical and machine-learning experiments.
- **Typesetting:** L<sup>A</sup>T<sub>E</sub>X.

## Selected Coursework

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- Calculus I and II (Honors);
- Linear Algebra I and II;
- Mathematical Analysis;
- Optimization;
- Probability I (Honors);
- Introduction to Stochastic Analysis
- Stochastic Differential Equations